STATISTICAL PHYSICS

Need for Statistical Mechanics:

Upto the begining of the eighteenth Century the physical phenomenon were explained by the ordinary laws of mechanics. But in certain cases particularly where the system contained a large no. of particles, these ordinary laws could not used either due to the mathematical complecations or due to their insufficiency for the interpretation of the observed phenomenon.

For example: We can obtain complete information concerning the motion of a mechanical system containing a large no. of particles by solving the equations of motion which are equal in no. to the degrees of freedom of the system.

Moreover the problem of an atom with two electrons presents a great mathematical complexity when the solution is tried with the use of ordinary laws of mechanics. It is thus impossible to solve the problem of a macroscopic body consisting of about 10^{23} atoms with their electrons per mole. Such problem have been successfully solved by the methods of statistical mechanics.

It is that branch of science which establishes the interpretation of the macroscopic behaviour of a system in terms of

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microscopic properties. It is not concerned with the actual motion of individual particles of a system but instead some average or most probable statistical properties of the system without going into the internal detail of the characteristics of its constituents.

In classical mechanics the state of a system of particles at any given instant of time is completely specified by the knowledge of position and momentum (or velocity) of each of its particles.

μ-Space & Γ-Space

The state of a single particle is completely defined by position co-ordinates x, y, z and momentum components p_x, p_y, p_z. We may now imagine a six dimensional space in which $dx.dy.dz.dp_x.dp_y.dp_z$ is a volume element and the positions of a particle in this space will be described by a set of six dimensional. space for a single particle is termed "phase space" or the "μ-space"

If the system contains a large no of particles such that the state of the system is represented by 'f' independent generalized position co-ordinates $q_1, q_2, q_3 \ldots q_f$ and 'f' generalyzed momenta co-ordinates, ~~then~~ ~~mathematically~~ the ~~and~~ 'f' ~~generalised~~ ~~momenta~~ ~~co-ordinates~~ $p_1, p_2 \ldots p_f$, then mathematically these '2f' combined position-momenta coordinates may be allowed to define a '2f' dimensional space called the

"Γ space" (which of course is a phase space)
An element of this space has a volume given by,

$$d\Gamma = (dq_1 \cdot dq_2 \cdot dq_3 \cdots dq_f)(dp_1 \cdot dp_2 \cdots dp_f)$$

The Γ-space is thus obviously due to superposition of many 'μ' spaces.
Again the dimension of the Γ-space will depend on the degrees of freedom of the system. An element of volume in the phase space is called a "cell".
Consider an element of volume $d\Gamma$ in the phase-space which is called a cell:
Then,

$$d\tau = dx \cdot dy \cdot dz \, dp_x dp_y \cdot dp_z \quad \text{— (1)}$$

According to Heisenbergs uncertainty principle we have

$$\left.\begin{aligned} dx\,dp_x &\geq \hbar \\ dy\,dp_y &\geq \hbar \\ dz\,dp_z &\geq \hbar \end{aligned}\right\}$$

Where, $\hbar = \frac{h}{2\pi}$; h being Plank's constant

Thus from eq^n (1): $d\tau = \hbar^3$

However on detailed calculation it is found that: $\boxed{d\tau = h^3}$